

SYNCKRO: Wearable Real-Time Multilingual Translator

1. Introduction

In today's globalized world, communication is essential, yet language barriers often limit effective interaction. To address this challenge, we developed **Synckro**, a wearable real-time multilingual translator that enables seamless communication between people speaking different languages.

Synckro is designed as a **wearable pendant**, allowing users to communicate naturally without holding any device. It captures speech, translates it in real time, and delivers output either publicly or privately.

2. Problem Statement

Existing translation systems have several limitations:

- Dependence on smartphones
- Lack of real-time conversational flow
- No privacy features
- No conversation tracking

These limitations reduce their effectiveness in real-world communication.

3. Objectives

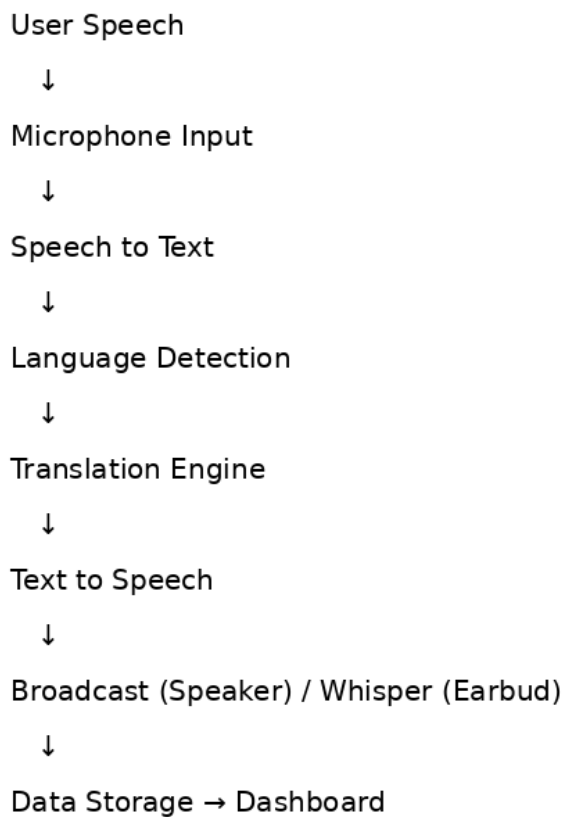
The objectives of this project are:

- To build a **real-time multilingual translation system**
- To design a **hands-free wearable interface**

- To implement **broadcast and whisper modes**
 - To support **online and offline translation**
 - To store conversations with **date, time, and language details**
 - To create a **dashboard interface** for conversation viewing
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4. System Architecture

The overall working of Synckro is shown below:



5. Working Methodology

The working flow of the system is illustrated below:

Start



User Speaks



Capture Audio



Convert Speech → Text



Detect Language



Translate Text



Convert Text → Speech



Select Mode



Broadcast → Speaker Output

Whisper → Earbud Output



Save Conversation



Display in Dashboard → End

6. Key Features

6.1 Real-Time Translation

The system translates speech instantly, allowing smooth conversation flow.

6.2 Multilingual Support

Supports multiple languages with automatic language detection.

6.3 Dual Output Modes

- **Broadcast Mode:** Output played aloud
- **Whisper Mode:** Output sent privately to earbud

6.4 Online and Offline Mode

- Online mode provides higher accuracy
- Offline mode works without internet

6.5 Conversation History

Stores all conversations with:

- Date
- Time
- Language

6.6 Dashboard Interface

Displays conversations in a chat-style format similar to messaging applications.

7. Hardware and Software Requirements

Hardware

- Microphone (input)
- Bluetooth earbud (output)

- Laptop (used as processing unit for prototype)

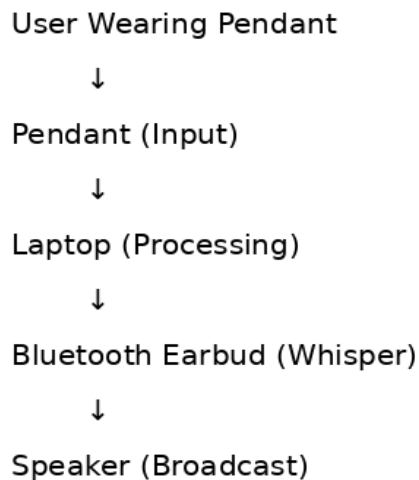
Software

- Python
 - Speech Recognition libraries (e.g., Vosk)
 - Translation APIs / models (Deep Translator, Argos Translate)
 - Text-to-Speech libraries (Pyttsx3)
 - JavaScript (dashboard)
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8. Prototype Design

The system is represented using a **wearable pendant prototype**. The processing unit is currently implemented externally using a laptop. In future, all components can be integrated into a compact wearable device.

9. Hardware Prototype Diagram



10. Data Flow Diagram

User Speech



Audio Input



Text Conversion



Translation



Audio Output



Storage Database



Dashboard UI

11. Applications

Synckro can be used in:

- Travel and tourism
 - Healthcare communication
 - Business meetings
 - Education
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12. Advantages

- Hands-free communication
 - Real-time translation
 - Multilingual support
 - Privacy via whisper mode
 - Conversation storage
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13. Limitations

- Slight delay in processing
 - Internet dependency for best accuracy
 - Hardware miniaturization not fully implemented
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14. Future Scope

- Full hardware integration into pendant
 - Improved offline accuracy
 - Integration with wearable devices
 - Enhanced noise filtering
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15. Conclusion

Synckro is an innovative system that enables seamless multilingual communication through real-time translation. By combining wearable design with intelligent software, it provides a practical and scalable solution for overcoming language barriers.

16. References

- [Speech Recognition Systems](#)
 - [Machine Translation Techniques](#)
 - [Text-to-Speech Technologies](#)
 - [Python Documentation](#)
 - [Online Translation APIs](#)
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